INCH-POUND

MIL-G-80088D
29 June 1988
SUPERSEDING
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30 March 1982

MILITARY SPECIFICATION

GRINDING MACHINES, SURFACE, RECIPROCATING TABLE, HORIZONTAL SPINDLE, POWER FEED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers horizontal spindle, reciprocating table, power feed grinders with equipment and accessories specified herein.

1.2 Classification. The grinding machine shall be of the following sizes. Worktables are available in various standard lengths as indicated. The size to be furnished shall be as specified (see 3.5 and 6.2.1).

Size	Worktable working surface, width (inches) x length (inches)
612	6 x 12/18
818	8 x 18/24
1024	10 x 24/30
1224	12 x 24/30/36/42/48/60/72/96
1424	14 x 24/36/48/60/72/96
1624	16 x 24/36/48/60/72/96/120/144/168/192/240
1848	18 x 48/60/72/84/96/120/144/168/192/240
2048	20 x 48/60/72/96/120/144/168/192/240
2460	24 x 60/72/96/120/144/168/192/216/240
3060	30 x 60/72/96/120/144/168/192/216/240
3660	36 x 60/72/96/120/144/168/192/216/240
4272	42 x 72/96/120/144/168/192/216/240
4872	48 x 72/96/120/144/168/192/216/240

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3415

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interferences.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

2.1.2 Other Government documents, drawings, and publications. The following Government document forms a part of this specification to the extent specified herein. The issues shall be those in effect on the date of the solicitation and may include title changes.

CODE OF FEDERAL REGULATIONS (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which are current on the date of the solicitation.

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 360.02 - Manual for Machine Tool Gearing.

AGMA 390.03 - Gear Classification, Materials and Measuring Methods for Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturers' Association, Standards Department, 1500 King Street, Suite 201, Alexandria, VA 22314-2717.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B5.1 - T-Slots, Their Bolts, Nuts and Tongues.

ANSI B7.1 - Safety Requirements for the Use, Care, and Protection of Abrasive Wheels.

ANSI B11.9 - Safety Requirements for the Construction, Care, and Use of Grinding Machines.

ANSI B74.2 - Specifications for Shapes and Sizes of Grinding Wheels, and Shapes, Sizes, and Identifications of Mounted Wheels.

ANSI B74.13 - Marking for Identifying Grinding Wheels and Other Bonded Abrasives.

ANSI/ISO 54 - Cylindrical Gears For General Engineering and for Heavy Engineering, Modules and Diametral Pitches of.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

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NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

ANSI/NEMA MG 1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

ANSI/NFPA 79 - Electrical Standard for Industrial Machinery.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Non-Government standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The machine shall be new (not a prototype) and one of the manufacturer's current production models capable of plunge, slot, and conventional surface grinding. The workpiece shall be supported on a reciprocating table and shall be ground using the periphery of a horizontally mounted grinding wheel. The machine shall have power operated axis drives and shall be either manually controlled by handwheels or pushbutton controls or automatically controlled by a feed cycle system set to predetermined parameters by the operator. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, and repair.

3.2.1 Design and construction dimensional systems. Unless otherwise specified (see 6.2.1), either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

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3.2.2 Measuring and indicating device calibrations. Measuring and indicating devices shall be graduated in US, SI, or dual (US and SI) units as specified by the procuring activity (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.2.1 Dials. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position. Dials indicating axis motion shall also meet the requirements of 3.4.10.6. All feed dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.2.2 Digital readout (DRO) system. Unless otherwise specified (see 6.2.1), a DRO system shall be installed on the Y axis (vertical head location) of travel. When specified (see 6.2.1), a DRO system shall also be provided on the Z axis (cross feed position) of travel. The DRO system shall consist of a position feedback device, an electronic display for each axis, and all related components, mounting hardware, and electrical equipment necessary to ensure complete operation of the system. The system shall provide a readily visible digital display over the full working range of each axis on which installed. The system resolution shall be 0.00005 inch. The system accuracy shall be equal to the system resolution, verified upon installation and calibration of each system on the machine. The repeatability shall be 0.0001 inch, measured over the full travel on each axis. Special requirements for the DRO system components shall be as specified (see 6.2.1). The DRO shall indicate axis location in both the US and SI units. An input power source fluctuation of $\pm$10 percent shall not adversely affect the DRO system operation.

3.2.2.2.1 Position feedback device. A position feedback device including all related components, hardware, and electrical equipment shall be provided for each axis having a readout system. The feedback device shall be of the independent type with a resolution equal to the resolution of the DRO unit. Each feedback device shall be installed with a cover or guard to prevent physical damage to any portion of the feedback device. The connecting cablefrom the position feedback device shall be electronically shielded to protect against electronic interference and shall be armored to protect against contamination and damage. The device shall provide axis travel feedback over the entire range of axis travel for each specified axis. Installation of the position feedback device shall not limit machine travel on any axis nor eliminate manual machine stops. Provisions shall be made to prevent damage to the feedback device in the event of overtravel. The feedback unit and related components shall be protected from contamination. Position feedback devices shall be easily and quickly interchangeable between like machine in any multiple machine order.

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3.2.2.2.2 Electronic display console. The electronic display console for the DRO system shall contain a universal digital display with individual illuminated digits of not less than 0.43 inch in height. Digits shall be high efficiency displays for high light ambient conditions with wide viewing angle and shall be readable at a distance of not less than 16 feet. The console shall display at least six digits per axis simultaneously. The display console shall also provide keypad entry, memory, absolute/incremental counting, data preset, power failure indicator, inch/metric conversion, and absolute zero recall. The electronic display console shall be installed so as to be readily accessible and easily read from the operator's work position. The console shall be mounted to withstand shock and vibration generated by the machine. The connecting cable shall be armored and shall be installed so as not to interfere with the normal operations of the machine. The cable shall be securely fastened to the structure of the machine in accordance with the requirements of the contract.

3.2.3 Reclaimed materials. The machine may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to means that the use of used or rebuilt products are allowed under this specification.

3.2.4 Energy efficiency. The machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency as governed by the developments available within the industry.

3.2.5 Controls. All mechanical, hydraulic, and pneumatic operating controls shall be located convenient to the operator's work station(s). All control switches shall be functionally grouped and clearly identified. An emergency control to stop all machine operations shall be provided.

3.2.6 Safety and health requirements. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 29 CFR 1910 and ANSI B11.9. If a conflict arises between 29 CFR 1910 and ANSI B11.9, 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified (see 6.2.1). Covers, guards, or other safety devices shall be provided for the point of operation and all other parts of the machine that present safety hazards. If specific point-of-guarding dependent on workpiece size and configuration is required, the procuring activity shall specify the exact configuration of the guard required (see 6.2.1).

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3.2.7 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.8 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine or its components.

3.2.9 Environmental protection. Under the operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation. Provisions should be made, as necessary, to provide for grinding dust collection during dry grinding (see 3.4.13.4).

3.2.10 Lubrication. All bearings (except sealed, permanently lubricated type), mating gears, and all other moving parts shall be provided means to ensure lubrication. Recirculating lubrication systems shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have at least a 24 hour capacity and means for determining fluid level. Means shall be provided to indicate a low oil condition. All oil holes, grease fittings, and filler caps shall be readily accessible.

3.2.11 Interchangeability. All parts subject to replacement shall be manufactured to definite dimensions and tolerances. Replacement parts (except for hand-fitted parts such as ways and boxed bearings) shall be interchangeable without requiring modification. Replacement parts are those which are subject to wear or failure such as gears, bearings, shafts, feed screws, drive belts, pumps, and seals. Electrical replacement parts shall include, but not be limited to, motors, relays, fuses, switches, magnetics, pushbuttons, light bulbs, braking mechanisms, and solid-state devices. Tooling and accessories normally supplied as standard items shall also be considered replacement parts. Foreign-sourced replacement parts shall be interchangeable with domestically supplied components or be readily available from a domestic source.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as weldings, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed to prevent loosening. Fastening devices subject to removal or adjustment shall not be permanently installed.

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3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Gears. All gears, shafts, and pinions of spindle and axis drive trains shall be designed and manufactured of a suitable ferrous material to meet or exceed the requirements of AGMA 360.02 and 390.03 for the US system or ANSI/ISO 54 for the SI system. The gears, shafts, and pinions shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears, shafts, and pinions in the drive train shall be hardened and ground steel. Working surface hardness shall be not less than Rockwell C-48.

3.3.8 Machine ways. Unless otherwise specified (see 6.2.1), all way surfaces shall be hardened, precision ground or hand scraped, and precision fit for smooth, accurate travel. Machine ways shall be protected regardless of machine component position to exclude contaminants.

3.3.9 Slides. All sliding members shall be provided with adjustable gibs and guides, or other means, as necessary, to maintain the alignment and accuracy tolerances required herein for the machine type specified.

3.3.10 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The machine shall consist of a base, saddle, worktable, wheelhead column(s), grinding wheelhead assembly, and feed mechanisms. Any additional components necessary for the operation of the machine shall be furnished in accordance with the manufacturer's standard practice.

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3.4.1 Base. The base shall be a box-type, stress-relieved casting of iron or iron alloy, ribbed and braced for rigidity. The base shall support the saddle, worktable, column, and wheelhead assembly. Means shall be provided to level and secure the machine to a flat mounting surface. On machines with a table width of less than 20 inches, longitudinal ways provided on the base shall be not less than 1-1/2 times the length of the worktable working surface. On machines with a table width of 20 inches or more, longitudinal ways provided on the base shall be not less than 1-1/2 time the length of the worktable working surface. Hydraulic and coolant reservoirs, as applicable, may be housed in the base or provided as separate external units. Access doors or removable plates shall be provided, as necessary, to service components housed in the base.

3.4.2 Saddle. The saddle shall be a casting of iron or iron alloy. The saddle shall traverse cross-axially (Z axis) either on guideways supported by the base or on guideways supported by the column. When the machine incorporates a saddle and worktable arrangement on the base, the top of the saddle shall have guideways to support longitudinal worktable movement. On machines with a table width of less than 20 inches, the saddle guideways shall be not less than 1-1/2 times the length of the worktable working surface and, on machines with a table width of 20 inches or more, the saddle guideways shall be not less than 2 times the length of the worktable working surface. When the machine incorporates a saddle and wheelhead assembly arrangement on the column, the saddle shall have guideways for facilitating cross-axial movement of the wheelhead assembly. Means shall be provided to prevent movement of the saddle after positioning during plunge or wheel form grinding. Stops shall be provided at each end of travel to prevent overtravel.

3.4.3 Worktable. The worktable shall be a casting of iron or iron alloy, ribbed to minimize distortion. The worktable shall travel longitudinally (X axis) on the guideways of the base or saddle, as applicable. The worktable shall be machined over the entire working surface. Means shall be provided to return spent coolant to its reservoir. Unless otherwise specified (see 6.2.1), the number of table T-slots shall be manufacturer's standard and shall meet the requirements of ANSI B5.1. Means shall be provided to prevent table overtravel. A T-slot or dog rack shall allow precise adjustment of reversing dogs. A grit deflector shall be provided.

3.4.4 Electromagnetic chuck. When specified (see 6.2.1), the machine shall be furnished with an electromagnetic chuck and all accessories necessary for installation and operation, including necessary equipment to magnetize and demagnetize the chuck. The chuck shall have full and variable holding power. The chuck shall be sealed to prevent fluid leakage into the coils or accumulation of moisture therein. The poles of the chuck shall be not more than 1/8 inch apart over the entire face of the chuck to ensure small parts as well as large parts will be held securely during grinding. The chuck shall operate from the same power source as the machine and shall be completely wired and connected to the main power panel on the machine. When specified (see 6.2.1), a safety interlock shall be provided to prevent spindle rotation or vertical wheelhead assembly travel unless the chuck is energized. The size of the chuck working surface and other special chuck features shall be as specified (see 6.2.1).

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3.4.5 Permanent magnetic chuck. When specified (see 6.2.1), the machine shall be furnished with a permanent magnetic chuck. The chuck shall be lever operated using no electric current. The size of the chuck working surface and other special features shall be as specified (see 6.2.1). When specified (see 6.2.1), a safety interlock shall be provided to prevent spindle rotation or vertical wheelhead assembly travel unless the chuck is energized.

3.4.6 Wheelhead column(s). The wheelhead assembly shall be mounted on a column(s). The column(s) shall be a one-piece, box-like casting, ribbed for rigidity and strength to minimize distortion, deflection, and vibration under normal workloads. The column(s) shall support and guide the wheelhead assembly or saddle through its full range of vertical adjustment on vertical (Y axis) guideways provided on the column(s). When a sliding saddle arrangement is supplied, guideways shall be provided for transverse movement of the wheelhead. When a sliding column(s) is supplied, its slides shall fit appropriate guideways on the base or column bed. Column surfaces, when fitted to the base surfaces, shall provide accuracies required herein. Stops shall be provided on the upper extreme of vertical travel to prevent overtravel.

3.4.7 Grinding wheelhead assembly. The grinding wheelhead assembly shall include a housing, a grinding wheel spindle, and spindle drive mechanism. The assembly shall travel vertically on the wheelhead column. When the machine incorporates cross-axial wheelhead movement, either the wheelhead shall traverse cross-axially on a saddle on guideways provided on the column or the wheelhead column shall traverse cross-axially on base guideways.

3.4.7.1 Grinding wheelhead spindle and spindle drive. The spindle shall be of steel alloy, supported and held in axial and radial alignment by precision, preloaded, antifriction ball or roller bearings. The spindle shall be dynamically balanced and provide vibration-free operation at all spindle speeds. Continuous or permanent lubrication to the spindle and bearings shall be provided. The spindle shall accept a driving mechanism at one end and a grinding wheel at the other end. The spindle shall accept taper-style wheel mounts or adapters for use with premounted and balanced grinding wheels. Spindle rotation shall be accomplished through a direct drive motor or a motor and belt and pulley arrangement. Motor speed shall be constant. When a belt and pulley arrangement is supplied, means shall be provided to adjust and maintain belt tension. All belts required for specified wheel speeds shall be furnished.

3.4.7.2 Grinding wheels. Grinding wheels shall conform to ANSI B74.2. The wheels shall be marked in accordance with ANSI B74.13 and shall meet the safety requirements established in ANSI B7.1. All grinding wheels shall be properly guarded. When specified (see 6.2.1), wheel guards shall have means to connect a dust extraction unit (see 3.4.13.4).

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3.4.8 Wheel truing device. Unless otherwise specified (see 6.2.1), the machine shall have an inplace diamond truing device arranged to infeed and dress the entire periphery of the grinding wheel. The dresser shall have either manual or power infeed as specified (see 6.2.1). The diamond shall be not less than one carat. The truing device shall permit wheel dressing without disturbing the workpiece set-up or changing the wheelhead vertical setting prior to wheel dressing. The device shall true wheels up to and including the size given in table I for the machine size specified.

3.4.9 Wheel forming device. When specified (see 6.2.1), the machine shall have a motor-driven wheel forming device. The device shall be easily mounted on the head or worktable in an area not normally used for grinding, be supported to withstand all forces encountered during operation, and be readily removable after use. Means shall be provided to interchange work rolls. The device shall form wheels up to and including the size given in table I for the machine size specified.

3.4.10 Feed mechanisms. Unless otherwise specified (see 6.2.1), the machine shall be provided with the following feed mechanisms.

3.4.10.1 Worktable longitudinal feed. The worktable shall be provided with automatic power operation for longitudinal movement. Table reversal shall be both manual and automatic through the use of adjustable stops. Table feeds shall be infinitely variable within the range specified in table I.

3.4.10.2 Cross feed. Cross feed shall be automatic by power to operate at each reversal of the worktable. Continuous feed shall be provided for wheel truing. The machine shall be equipped with adjustable cross feed stops and a fully automatic mechanism for reversing the direction of cross feed travel at any predetermined point. The cross feed settings shall be infinite within the range specified in table I for the machine size specified.

3.4.10.3 Wheelhead power positioning. The machine shall be provided with a power operated drive for raising and lowering the wheelhead on the column for rapid positioning and wheel dressing.

3.4.10.4 Automatic grinding wheel down feed. When specified (see 6.2.1), an automatic power down feed cycle system shall be provided. For slot and plunge grinding, down feed actuation shall be controlled by table reversals. For conventional surface grinding, the cycle system shall automatically lower the wheel at each cross feed reversal. The operator shall set the amount of automatic down feed to a predetermined stop, ranging from 0.001 inch to 0.040 inch, in increments from 0.00025 inch or less to 0.001 inch or more. When specified (see 6.2.1), the optional down feed increments of 0.0001 inch to 0.001 inch shall be provided in lieu of the standard increments. It shall be possible to alter the feed rate during the automatic down feed cycle without changing the finished part size. Upon reaching the preset down feed limit, a minimum of three spark out passes at zero down feed shall be made. When specified (see 6.2.1), after the spark out passes have been completed, the wheel shall automatically retract vertically to load position or other position as specified and stop rotating.

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3.4.10.5 Grinding wheel wear compensation. When specified (see 6.2.1), means shall be provided to automatically compensate for wheel wear during the automatic grinding wheel down feed cycle.

3.4.10.6 Manual feeds. Means shall be provided to manually control the vertical movement of the grinding wheelhead assembly. The handwheel shall be graduated in increments of 0.001 inch or less, directly readable on a dial or vernier. A fine feed dial shall also be provided for down feed, graduated in increments of not greater than 0.0001 inch. Unless otherwise specified (see 6.2.1), a cross feed handwheel shall be provided for the worktable or saddle, as applicable, for manual set-up, inspection, and positioning. Cross feed dial increments shall be 0.001 inch or less. Means shall be provided to prevent simultaneous engagement of handwheels and power feed.

3.4.11 Coolant system. Unless otherwise specified (see 6.2.1), a coolant system shall be provided. The coolant system shall include a sump or reservoir, a power driven pump, and all necessary piping. The sump or reservoir shall have sufficient capacity to permit full flow of coolant. The system shall have means for draining, cleaning, and determining the fluid level. Means shall be provided to permit the operator to direct and control the amount of coolant. An interlock shall be provided to prevent the use of a dust collection system (see 3.4.13.4) during wet grinding. Means shall be provided to select either wet grinding (coolant) or dry grinding with dust extraction.

3.4.12 Hydraulic system. The hydraulic system shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided in the high pressure line to prevent damage to the components. A filter system shall be provided to ensure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level, provisions for draining, and a cleanout access plate if the reservoir is not removable.

3.4.13 Electrical system. The electrical system shall conform to ANSI/NFPA 79. Unless otherwise specified (see 6.2.1), each machine shall draw all of its electrical power from a single 230/460-volt, 3-phase, 60-hertz (Hz) circuit, maximum line voltage fluctuation of 10 percent rated voltage, having a fused safety disconnect switch or circuit breaker. Unless otherwise specified (see 6.2.1), the machine shall be initially wired for operation on 460 volts.

3.4.13.1 Motors. Dual voltage motors shall be provided. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2.1), alternating-current (AC) motors shall be designed to operate on 60-Hz. Each motor enclosure shall meet the requirements for a dripproof enclosure. All motors shall meet the requirements of ANSI/NEMA MG-1. Each motor shall have an identification plate to identify manufacturer, model identification, serial number, voltage, amperage, horsepower, phase, and frequency.

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3.4.13.2 Worklight. Unless otherwise specified (see 6.2.1), a worklight shall be provided. The light shall have a protective shield and an on-off switch and shall be adjustable to reflect maximum light on the work area. If a built-in lighting system is provided, the adjustable light described above shall not be required.

3.4.13.3 Hour meter. When specified (see 6.2.1), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 9,999 hours in increments of not greater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.4.13.4 Dust extractor. Unless otherwise specified (see 6.2.1), a dust extractor and all necessary connections shall be provided to be used during dry grinding (an individual dust extraction unit is not necessary if a central dust extraction system is available). The dust extractor shall be connected to the machine's electrical system with start/stop pushbuttons conveniently located on the control panel. The dust extractor shall have the special features and be of the efficiency rating as specified (see 6.2.1). An interlock shall be provided to prevent wet grinding (see 3.4.11) while using the dust extractor. Means shall be provided to select either wet grinding (coolant) or dry grinding with dust extraction.

3.4.13.5 Electrical load indicator. When specified (see 6.2.1), the machine shall have a meter to indicate the electrical load on the spindle drive motor. The meter shall be current actuated. The dial shall indicate either amperes or percent of full-load current and shall be clearly marked to indicate the rated reading for continuous operation of the machine. The accuracy of the meter shall be within 2 percent of full-rated load. The meter shall be clearly readable from the operator's position.

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the machine shall be not less than specified in table I for the machine size specified.

3.6 Alignment tolerances. Each machine shall meet the alignment tolerances specified in table II when tested using the procedures given in 4.7.2.

3.7 Performance. The machine shall be capable of meeting the performance requirements given in 4.7.3.

3.7.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

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3.8 Standard equipment. Unless otherwise specified (see 6.2.1), the following equipment shall be furnished as standard with each machine:

a. One set of special wrenches, operating, and repair tools normally furnished by the manufacturer.

b. One set of straight grinding wheels normally furnished by the manufacturer (not less than two).

c. One extra set of wheelhead belts, if applicable.

3.9 Optional equipment. The following equipment and accessories are commercially available and are considered optional. When required, optional equipment, quantity and description, shall be as specified (see 6.2.1).

a. Wheel balancing device.

b. Tool tray.

c. Additional grinding wheels: shank size, number of each wheel size, type, dimensional characteristics, and composition of each wheel, as specified (see 6.2.1).

d. Index centers of the specified size for grinding taps, reamers, and cutters (see 6.2.1).

e. Sine plate(s).

f. Vise(s).

3.10 Repair parts. Required repair parts shall be furnished as specified (see 6.2.1). All repair parts shall be assigned a part number identified in a repair parts manual.

3.11 Marking on instruments, control panels, and plates. All words on instruments, control panels, and plates shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background. All plates shall be corrosion resistant.

3.11.1 Lubrication plate. A lubrication plate shall be permanently and securely attached to each machine. Unless otherwise specified (see 6.2.1), the plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity
Military or Federal specification number for each lubricant
Size and type of each lubricant filter

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3.11.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2.1), the nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model number of the basic standard machine and a suffix identified in the manufacturer's permanent records.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.12 Technical data. When technical data is required, it shall be furnished in accordance with the requirements of DD Form 1423. All technical data furnished shall be written in the English language.

3.13 Workmanship. Workmanship of the machine shall meet all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements (see 6.6).

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

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a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2.1), all inspections, tests, and examinations shall be performed in an indoor facility with ambient conditions of 41°F to 104°F and 20 percent to 95 percent relative humidity.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. The machine shall pass the first article examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.6, the tests in 4.7, and the inspection in 4.8. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Examination. The machine shall be examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Operational test. The machine shall be operated at no load for not less than one hour. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.2 Alignment tolerance test. Each machine shall be tested to determine compliance with the alignment tolerance requirements of 3.6. The alignment tolerance test shall be performed prior to and subsequent to the performance test(s) given in 4.7.3 through 4.7.3.3. Any machine adjustments required to comply with 3.6 during or after the performance test shall be cause for rejection.

4.7.3 Performance tests. The machine shall be subjected to the following performance tests. At the option of the procuring agency, each machine shall be subjected to tests in accordance with specific requirements designated by the activity in lieu of or in addition to the following tests (see 6.2.1). All performance tests shall be completed without making any adjustments to or performing any maintenance on the machine.

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4.7.3.1 Block grinding test. The machine shall be tested to determine its ability to surface grind. The machine shall grind the top and bottom surfaces of five test blocks, one located at each corner and one in the center of the worktable. The number of test blocks used for this test may be increased for machines with worktables over 48 inches in length (see 6.2.1). The test blocks shall be the manufacturer's standard test blocks. For machines with worktables 36 inches or less in length, the blocks shall have a 4 square inch surface area and be 1 inch thick (minimum). For machines with worktables longer than 36 inches, the blocks shall have a 9 square inch surface area and be 1 inch thick (minimum). The method and grinding wheels used to perform this test shall be at the option of and supplied by the manufacturer. The machine shall surface grind the top and bottom of the five test blocks at any location on the table working surface parallel to within 0.0001 inch per foot and flat to within 0.0001 inch per foot. The five test blocks shall be ground equal in size to within 0.0001 inch per foot.

4.7.3.2 Slot grinding test. The machine shall grind a slot in a test part of SAE 01 tool steel (59-60 Rc). The slot shall be 1/2 inch deep and have a length 4 inches less than the worktable length or 24 inches, whichever is less. The slot may be roughed out by any method chosen by the supplier. The slot shall be ground at any location on the worktable with the two sides parallel to within 0.0001 inch per foot and each side square with the bottom in a vertical plane to within 0.0001 inch.

4.7.3.3 Surface finish grinding test. The machine shall be tested to determine its ability to finish grind a workpiece. The test blocks used for the block grinding test (see 4.7.3.1) shall be used. The machine shall finish grind the top and bottom surfaces of the test blocks to a surface finish of not rougher than 10 microinch. A grinding wheel change may be made prior to the start of the surface finish grinding test, but not during the execution of the test.

4.7.4 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.10 using the procedures given in MIL-STD-462.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

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6. NOTES

6.1 Intended use. The machines covered by this specification are for use in tool and die, maintenance, and production shops for rough and precision finish grinding, slot, or form grinding operations to be performed on a wide variety of materials.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Size required (see 1.2).

c. If machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

d. System of units for measuring and indicating devices (see 3.2.2).

e. Digital readout system, if not required (see 3.2.2.2).

f. Specify DRO system on Z axis (cross feed position), if required (see 3.2.2.2).

g. Specify any special DRO system requirements, if required (see 3.2.2.2).

h. Additional safety and health requirements, if required (see 3.2.6).

i. Specify configuration of additional point-of-operation guarding, if required (see 3.2.6).

j. Painting, if different (see 3.3.5).

k. Way surfaces, if different (see 3.3.8).

l. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.10).

m. Table T-slots, if different (see 3.4.3).

n. Electromagnetic chuck, if required (see 3.4.4).

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o. Safety interlock on electromagnetic chuck, if required (see 3.4.4).

p. Electromagnetic chuck working surface size and special features, if required (see 3.4.4).

q. Permanent magnetic chuck, if required (see 3.4.5).

r. Permanent magnetic chuck working surface size and special features, if required (see 3.4.5).

s. Safety interlock on permanent magnetic chuck, if required (see 3.4.5).

t. Wheel guards with dust extractor connections, if required (see 3.4.7.2).

u. Wheel truing device, if not required (see 3.4.8).

v. Dresser manual or power infeed, as specified (see 3.4.8).

w. Wheel forming device, if required (see 3.4.9).

x. Feed mechanisms, if different (see 3.4.10).

y. Automatic grinding wheel down feed cycle system, if required (see 3.4.10.4).

z. Optional down feed increments of 0.0001 inch to 0.001 inch, if required (see 3.4.10.4).

aa. Automatic retraction to load position or other position, as specified, if required (see 3.4.10.4).

bb. Grinding wheel wear compensation, if required (see 3.4.10.5).

cc. Cross feed handwheel, if not required (see 3.4.10.6).

dd. Coolant system, if not required (see 3.4.11).

ee. Electrical system, if different (see 3.4.13).

ff. Initial input voltage, if different (see 3.4.13).

gg. Motors, if different (see 3.4.13.1).

hh. Worklight, if not required (see 3.4.13.2).

ii. Hour meter, if required (see 3.4.13.3).

jj. Dust extractor, if not required (see 3.4.13.4).

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kk. Special dust extractor features and efficiency rating, if required (see 3.4.13.4).

ll. Electrical load indicator, if required (see 3.4.13.5).

mm. Size and capacity, if different (see 3.5).

nn. Additional performance features, if required (see 3.7.1).

oo. Standard equipment, if different (see 3.8).

pp. Specify and fully describe optional equipment, if required (see 3.9).

qq. Specify additional grinding wheels (shank size, number of each wheel size, type, dimensional characteristics, and composition of each wheel), if required (see 3.9.c).

rr. Specify size(s) of index centers, if required (see 3.9.d).

ss. Specify applicable repair parts, if required (see 3.10).

tt. Lubrication plate, if different (see 3.11.1).

uu. Nameplate, if different (see 3.11.2).

vv. Inspection conditions, if different (see 4.3).

ww. First article inspection, if different (see 4.4).

xx. Quality conformance inspection, if different (see 4.5).

yy. Performance tests, if different (see 4.7.3).

zz. Specify additional number of test blocks for performance test, if required (see 4.7.3.1).

aaa. If preservation, packing, and marking in accordance with MIL-M-18058 is required, specify level required and specification number (see 5.1).

6.2.2 Contract data requirements. Required technical data (operating manuals, parts, lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports) should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. Specific instructions regarding arrangements for examination, test, and approval of the first article are given in the contract.

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6.4 Cross-reference of classification changes. Machine classification changes from the previous issue of this specification are as follows:

MIL-G-80088C	MIL-G-80088D
Size 612	Size 612
Size 818	Size 818
Size 1024	Size 1024
Size 1224	Size 1224
Size 1424	Size 1424
Size 1624	Size 1624
Size 1836	Deleted
New	Size 1848
Size 2036	Deleted
New	Size 2048
Size 2448	Deleted
New	Size 2460
Size 3060	Size 3060
New	Size 3660
Size 3672	Size 3660 (optional table length)
Size 4272	Size 4272
Size 4872	Size 4872

6.5 Changes from previous issues. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.6 Inspection facilities. The contractor must identify in his response to the solicitation the location where inspection and tests are to be performed. Unless otherwise provided for in the contract, all inspections will be performed within the limits of the continental U.S.

6.7 Training. Training required by the procuring activity shall be provided as specified in the contract.

6.8 Warranty. Warranty requirements shall be as specified by the procuring activity in the contract.

6.9 Subject term (keyword) listing.

Automatic
Grinder, surface
Machine tool, grinding, surface
Magnetic chuck, electromagnetic, permanent
Reciprocating table
Spindle, horizontal
Surface grinder
Wheel, abrasive, grinding

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TABLE I. Sizes and capacities.1/

Characteristic	Size 612	Size 818	Size 1024	Size 1224	Size 1424
Capacity:					
Worktable surface, inches	6 x 12	8 x 18	10 x 24	12 x 24	14 x 24
Optional worktable length, inches	18	24	30	30/36/42/48/ 60/72/96	36/48/60/ 72/96
Table top to bottom of new standard wheel, inches	12	12	12	15	15
Cross feed travel, inches	6-3/4	9	11-1/4	13-1/4	15-1/4
Cross feed range, inches	1/10 to 1/4	1/10 to 1/4	1/10 to 3/8	1/10 to 3/8	1/10 to 1/2
Table longitudinal travel, inches 2/	12-3/4	22	29	29	35
Wheelhead vertical travel, inches	13	15	15	17	18
Standard grinding wheel size (new), diameter x thickness, inches 2/	7 x 1/2	10 x 1/2	12 x 1	12 x 1	14 x 1
Feeds and speeds:					
Spindle speed, RPM	3600	2420	1760	1725	1750
Table longitudinal feed, FPM	10 - 70	10 - 100	10 - 100	10 - 100	10 -100
Cross feed, IPM	0 - 75	0 - 75	10 - 75	10 - 100	10 -120
Rapid vertical positioning, IPM	0 - 10	0 - 10	0 - 10	0 - 10	0 - 10
Motor horspowers (minimum):					
Spindle 2/	1	2	5	5	7-1/2
Optional spindle	—	—	--	--	15
Hydraulic pump	1/2	1-1/2	1-1/2	3	3
Power elevation	1/4	1/4	1/4	1/4	1/4

NOTE 1/: Machine sizes and capacities shall be not less than the stated requirements of the size ordered. When a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater. Machine physical size shall not be greater than the requirements of the next larger size shown in above table. When the largest size machine shown in above table is ordered, the size and capacity of the machine offered shall not exceed the stated physical size requirements by more than 10 percent.

NOTE 2/: Longitudinal travel, standard wheel size, and spindle horsepower may vary for optional table sizes.

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TABLE I. Sizes and capacities - Continued.1/

Characteristic	Size 1624	Size 1848	Size 2048	Size 2460
Capacity:				
Worktable surface, inches	16 x 24	18 x 48	20 x 48	24 x 60
Optional worktable length, inches	36/48/60/72/96/120/144/168/192/240	60/72/84/96/120/144/168/192/240	60/72/96/120/144/168/192/240	72/96/120/144/168/192/216/240
Table top to bottom of new standard wheel, inches	17	18	18	25
Cross feed travel, inches	17	18	20	24
Cross feed range, inches	1/10 to 1/2	1/32 to 1-1/2	1/32 to 1-1/2	1/32 to 2
Table longitudinal travel, inches 2/	35-1/2	60	60	70
Wheelhead vertical travel, inches	18	18	18	25
Standard grinding wheel size (new), diameter x thickness, inches 2/	14 x 1	20 x 3	20 x 3	20 x 4
Feeds and speeds:				
Spindle speed, RPM	1200	1200	1200	1200
Table longitudinal feed, FPM	10 - 100	20 - 100	10 - 100	10 - 100
Cross feed, IPM	1 - 120	0 - 220	0 - 220	0 - 220
Rapid vertical positioning, IPM	0 - 10	0 - 10	0 - 10	0 - 10
Motor horsepowers (minimum):				
Spindle 2/	7-1/2	15	15	30
Optional spindle	15	--	--	--
Hydraulic pump	3	7-1/2	7-1/2	15
Power elevation	1/4	1/4	1/4	1/4

NOTE 1/: Machine sizes and capacities shall be not less than the stated requirements of the size ordered. When a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater. Machine physical size shall not be greater than the requirements of the next larger size shown in above table. When the largest size machine shown in above table is ordered, the size and capacity of the machine offered shall not exceed the stated physical size requirements by more than 10 percent.

NOTE 2/: Longitudinal travel, standard wheel size, and spindle horsepower may vary for optional table sizes.

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TABLE I. Sizes and capacities - Continued.1/

Characteristic	Size 3060	Size 3660	Size 4272	Size 4872
Capacity:				
Worktable surface, inches	30 x 60	36 x 60	42 x 72	48 x 72
Optional worktable length, inches	72/96/120/ 144/168/192/ 216/240	72/96/120/ 144/168/192/ 216/240	96/120/144/ 168/192/216 240	96/120/144/ 168/192/216 240
Table top to bottom of new standard wheel, inches	25	25	36	36
Cross feed travel, inches	30	36	42	46
Cross feed range, inches	1/4 to 2	1/4 to 2	1/4 to 2	1/4 to 3
Table longitudinal travel, inches 2/	70	70	74	98
Wheelhead vertical travel, inches	26	37	37	37
Standard grinding wheel size (new), diameter x thickness, inches 2/	20 x 4	20 x 4	20 x 4	20 x 6
Feeds and speeds:				
Spindle speed, RPM	1200	1200	1200	1200
Table longitudinal feed, FPM	10 - 100	10 - 100	10 - 100	15 - 100
Cross feed, IPM	0 - 220	0 - 220	0 - 220	0 - 220
Rapid vertical positioning, IPM	0 - 18	0 - 18	0 - 18	0 - 18
Motor horsepowers (minimum):				
Spindle 2/	30	30	30	50
Optional spindle	--	--	50	--
Hydraulic pump	15	15	15	25
Power elevation	1/4	1/4	2	2

NOTE 1: Machine sizes and capacities shall be not less than the stated requirements of the size ordered. When a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater. Machine physical size shall not be greater than the requirements of the next larger size shown in above table. When the largest size machine shown in above table is ordered, the size and capacity of the machine offered shall not exceed the stated physical size requirements by more than 10 percent.

NOTE 2/: Longitudinal travel, standard wheel size, and spindle horsepower may vary for optional table sizes.

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TABLE II. Alignment and accuracy requirements.

Test to be applied	Permissible tolerance
Worktable top flat lengthwise, crosswise, and diagonally (no twist permitted)	0.0002"/FT
Rise and fall of worktable in its longitudinal movement	0.0001"/FT
Worktable T-slots parallel with worktable longitudinal movement	0.0005"/FT
Surface of worktable parallel with transverse movement	0.0003"/18"
Squareness of longitudinal and cross feed travel	0.001"/FT
Periphery runout of wheel spindle - for machines up to 30" in table length - for machines 30" and over in table length	 0.0001" TIR 0.0004" TIR
Wheel spindle axial slip - for machines up to 30" in table length - for machines 30" and over in table length	 0.0001" TIR 0.0004" TIR
Wheel spindle axis square to worktable surface (using a 6", 180 radius sweep in a vertical plane)	0.0005" TIR
Wheel spindle axis parallel to worktable surface	0.0006"/18"
Wheel spindle axis square with worktable longitudinal travel	0.0004"/FT
Vertical adjustment of wheelhead assembly square with worktable surface in transverse plane (using two readings 90 apart)	0.001"/FT

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Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Army - AR, AV
Navy - YD
Air Force - 84

User activities:
Army - SM, AR
Navy - MC

Preparing activity:
DLA - IP

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-80088D	Grinding Machines, Surface, Recip Table, Horiz, Power Feed

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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